

Prompt Engineering Internship – Task 1 Submission Report

Prompt Basics: Zero-Shot vs. Few-Shot Showdown

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Task: Zero-Shot vs. Few-Shot Prompting Comparison

1. Objective

The objective of this task was to compare **Zero-Shot Prompting** and **Few-Shot Prompting** by classifying customer support messages into three categories:

- Complaint
- Question
- Praise

The prompts were tested using three different Large Language Models:

- ChatGPT
 - Claude
 - Gemini
-

2. Dataset

The following 10 customer support messages were used for evaluation.

Message No.	Customer Message	Correct Label
1	My order arrived two days late and the package was damaged.	Complaint
2	How can I reset my password?	Question
3	Your customer support was amazing. Thank you!	Praise

4	The app keeps crashing every time I open it.	Complaint
5	Can I change my delivery address after placing an order?	Question
6	I love the new update. It is much faster now.	Praise
7	I was charged twice for the same purchase.	Complaint
8	Do you offer international shipping?	Question
9	Excellent service! I will recommend you to everyone.	Praise
10	The website is extremely slow today.	Complaint

3. Zero-Shot Prompt

You are a customer support classification assistant.

Classify each customer message into exactly one of these categories:

- Complaint
- Question
- Praise

Return your answer in the following table format.

Message Number | Category

Messages:

1. My order arrived two days late and the package was damaged.
2. How can I reset my password?
3. Your customer support was amazing. Thank you!
4. The app keeps crashing every time I open it.
5. Can I change my delivery address after placing an order?
6. I love the new update. It is much faster now.
7. I was charged twice for the same purchase.
8. Do you offer international shipping?

9. Excellent service! I will recommend you to everyone.

10. The website is extremely slow today.

4. Few-Shot Prompt

You are a customer support classification assistant.

Classify each customer message into one of the following categories:

- Complaint
- Question
- Praise

Below are some examples.

Example 1

Message:

"My refund has not arrived."

Category:

Complaint

Example 2

Message:

"How can I update my account information?"

Category:

Question

Example 3

Message:

"I really love your service."

Category:

Praise

Now classify the following customer messages.

1. My order arrived two days late and the package was damaged.

2. How can I reset my password?

3. Your customer support was amazing. Thank you!
4. The app keeps crashing every time I open it.
5. Can I change my delivery address after placing an order?
6. I love the new update. It is much faster now.
7. I was charged twice for the same purchase.
8. Do you offer international shipping?
9. Excellent service! I will recommend you to everyone.
10. The website is extremely slow today.

Return your answer in the following format.

Message Number | Category

5. Correct Classification

Message No.	Category
1	Complaint
2	Question
3	Praise
4	Complaint
5	Question
6	Praise
7	Complaint
8	Question
9	Praise
10	Complaint

6. Results

Zero-Shot Results

AI Tool	Correct	Accuracy
ChatGPT	10/10	100%
Claude	10/10	100%
Gemini	10/10	100%

Few-Shot Results

AI Tool	Correct	Accuracy
ChatGPT	10/10	100%
Claude	10/10	100%
Gemini	10/10	100%

7. Comparison Table

AI Tool	Zero-Shot	Few-Shot
ChatGPT	10/10 (100%)	10/10 (100%)
Claude	10/10 (100%)	10/10 (100%)
Gemini	10/10 (100%)	10/10 (100%)

8. Discussion

In this experiment, both Zero-Shot and Few-Shot prompting achieved **100% accuracy** across ChatGPT, Claude, and Gemini. The customer support messages were clear and easy to classify, allowing all three models to correctly identify each category without difficulty.

Although the measured accuracy was the same, Few-Shot Prompting remains an important prompting technique. By providing labeled examples before the task, it helps the model better understand the expected output format and decision pattern. This becomes especially valuable when working with ambiguous, complex, or domain-specific data where zero-shot prompting may produce inconsistent results.

9. Conclusion

This task demonstrated the practical difference between Zero-Shot and Few-Shot prompting. On a simple classification task, both approaches performed equally well, achieving perfect accuracy across all three AI models. However, Few-Shot Prompting offers additional guidance through examples, making it a more reliable approach for challenging or less obvious classification tasks. Overall, this exercise highlighted the importance of selecting the appropriate prompting strategy based on the complexity of the problem.
